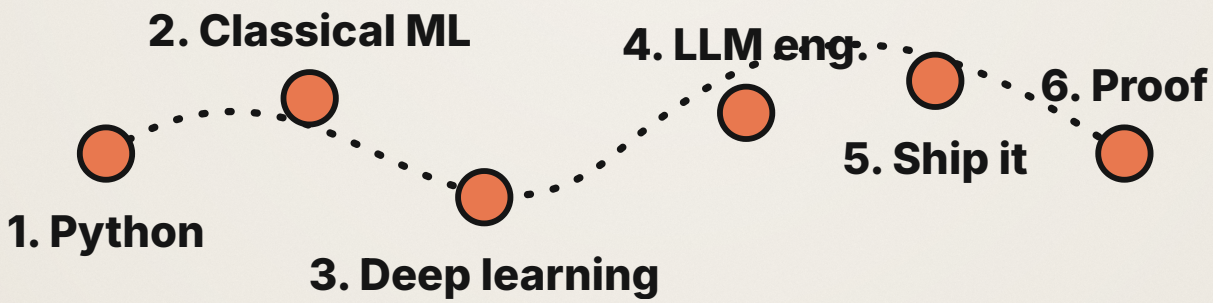
AI Engineer 2026

Skip the 50 courses. Six steps, in this order, with the free resources I'd use for each one.



You don't need a PhD

Build apps, not notebooks

Ship it

STEP 1 / 6 · WEEKS 1–6

1 Python + just enough maths

Everything else is written in Python. The maths is only what you need to read a model, not a PhD.

WHAT TO LEARN

- Functions, classes, async, virtual envs and pip
- NumPy and pandas basics
- Vectors and matrices, probability and distributions, derivatives and gradients

FREE RESOURCES

Python Tutorial Python docs
`docs.python.org/3/tutorial`

CS50's Introduction to Programming with Python Harvard, free
`cs50.harvard.edu/python`

Automate the Boring Stuff with Python Al Sweigart, free online
`automatetheboringstuff.com`

Essence of Linear Algebra 3Blue1Brown
`www.3blue1brown.com/topics/linear-algebra`

Essence of Calculus 3Blue1Brown
`www.3blue1brown.com/topics/calculus`

Statistics and Probability Khan Academy
`www.khanacademy.org/math/statistics-probability`

Mathematics for Machine Learning Deisenroth et al., free book
`mml-book.github.io`

Done when You can write an async script that calls an API and saves the results, and explain a dot product and a gradient in plain words.

2

STEP 2 / 6 · WEEKS 7–12

Classical ML

This is where you learn why models fail: overfitting, leakage, bad metrics. That instinct carries into everything after.

WHAT TO LEARN

- Regression and classification
- Train / test split, cross-validation
- Overfitting and how to spot it
- Evaluation metrics: accuracy, precision, recall, F1

FREE RESOURCES

scikit-learn User Guide scikit-learn
scikit-learn.org/stable/user_guide.html

Kaggle Learn: Intro + Intermediate ML Kaggle, free
www.kaggle.com/learn

Machine Learning Specialization Andrew Ng · DeepLearning.AI and Stanford, free to audit
www.coursera.org/specializations/machine-learning-introduction

Done when You trained a model, compared train vs test scores, and can say why it overfit and what you changed.

3

STEP 3 / 6 · WEEKS 13–20

Deep learning

Build one neural network from scratch, once. Then understand transformers, because everything today runs on them.

WHAT TO LEARN

- Tensors, autograd and the training loop in PyTorch
- A small network from scratch
- Attention and the transformer block

FREE RESOURCES

PyTorch Tutorials PyTorch
pytorch.org/tutorials

Neural Networks: Zero to Hero Andrej Karpathy
karpathy.ai/zero-to-hero.html

Practical Deep Learning for Coders fast.ai
course.fast.ai

The Illustrated Transformer Jay Alammar
jalammar.github.io/illustrated-transformer

Attention Is All You Need Vaswani et al., 2017
arxiv.org/abs/1706.03762

Done when You wrote a training loop yourself and can explain attention on a whiteboard.

4

STEP 4 / 6 · WEEKS 21–28

LLM engineering

This is where the jobs are. Most AI engineering work is building reliable products on top of models, not training them.

WHAT TO LEARN

- Prompting and model APIs
- Embeddings and a vector database
- RAG (retrieval-augmented generation)
- Fine-tuning when prompting isn't enough
- Evals, so you know when it got worse

FREE RESOURCES

Prompt engineering guide Anthropic docs
`docs.anthropic.com/en/docs/build-with-claude/prompt-engineering/overview`

OpenAI API docs OpenAI
`platform.openai.com/docs`

LLM Course Hugging Face, free
`huggingface.co/learn/llm-course`

Short courses DeepLearning.AI, free
`www.deeplearning.ai/short-courses`

Sentence Transformers (embeddings) SBERT
`www.sbert.net`

pgvector (vectors in Postgres) GitHub
`github.com/pgvector/pgvector`

PEFT (efficient fine-tuning) Hugging Face docs
`huggingface.co/docs/peft`

Your AI Product Needs Evals Hamel Husain
`hamel.dev/blog/posts/evals`

Done when A RAG app answers questions over your own documents, and an eval set tells you when a change made it worse.

5

STEP 5 / 6 · WEEKS 29–34

Ship it

An AI engineer who can't ship is just a researcher. Deploy it, watch it, know what it costs.

WHAT TO LEARN

- Docker
- FastAPI
- Deploying to a cloud
- Monitoring: logs, latency, errors
- Cost tracking per request

FREE RESOURCES

Docker: Get Started Docker docs
`docs.docker.com/get-started`

FastAPI Tutorial FastAPI docs
`fastapi.tiangolo.com/tutorial`

Made With ML (MLOps) Goku Mohandas
`madewithml.com`

Full Stack Deep Learning FSDL
`fullstackdeeplearning.com`

Langfuse (open-source LLM tracing and costs) Langfuse docs
`langfuse.com/docs`

Done when Your app runs in a container at a public URL, and you can see each request's latency and cost.

6

STEP 6 / 6 · START NOW, KEEP GOING

Proof of work

Three real projects beat fifty certificates. Write about what broke: that's your resume.

PROJECT IDEAS

- RAG chatbot over a docs site you actually use
- A fraud detector with an honest eval report (the Kaggle dataset below)
- An LLM eval harness that scores prompts or models on your own test set

FREE RESOURCES

GitHub: Get started GitHub docs
docs.github.com/en/get-started

Credit Card Fraud Detection dataset Kaggle
www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Done when Each repo has a README with a demo, and each has a short post on what broke and how you fixed it.

YOUR CHECKLIST

Tick one off, *then move on*

- [ ] **1. Python + just enough maths**
  You can write an async script that calls an API and saves the results, and explain a dot product and a gradient in plain words.

- [ ] **2. Classical ML**
  You trained a model, compared train vs test scores, and can say why it overfit and what you changed.

- [ ] **3. Deep learning**
  You wrote a training loop yourself and can explain attention on a whiteboard.

- [ ] **4. LLM engineering**
  A RAG app answers questions over your own documents, and an eval set tells you when a change made it worse.

- [ ] **5. Ship it**
  Your app runs in a container at a public URL, and you can see each request's latency and cost.

- [ ] **6. Proof of work**
  Each repo has a README with a demo, and each has a short post on what broke and how you fixed it.

Three projects on GitHub + a write-up of what broke

= *your resume.*

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